TINY PROJECT

Java Access Modifiers

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Project Topic	Demonstration of Public, Private, Protected and Default Member Access

Java Programming Project

1. Project Overview

This tiny project demonstrates how Java access modifiers control the visibility of class members. The application uses classes placed in different packages to demonstrate public, private, protected and default access.

2. Objective

• Understand the purpose of Java access modifiers.

• Demonstrate public, private, protected and default members.

• Use classes from different packages.

• Understand how inheritance affects protected access.

• Observe which members can and cannot be accessed.

3. Project Structure

```
TinyProject/
■
■■■ src/
■   ■■■ package1/
■   ■   ■■■ Parent.java
■   ■
■   ■■■ package2/
■   ■   ■■■ Child.java
■   ■
■   ■■■ Main.java
■
■■■ README.txt
```

package1 contains Parent, **package2** contains Child, and Main.java is outside both packages. Child extends Parent to demonstrate protected access.

4. Source Code

File 1: Parent.java

```
package package1;

public class Parent {

    public int publicVar = 10;
    private int privateVar = 20;
    protected int protectedVar = 30;
    int defaultVar = 40;

    public void display() {
        System.out.println("Inside Parent Class");
        System.out.println("--------------------");
        System.out.println("Public Variable    : " + publicVar);
        System.out.println("Private Variable   : " + privateVar);
        System.out.println("Protected Variable : " + protectedVar);
        System.out.println("Default Variable   : " + defaultVar);
    }
}
```

File 2: Child.java

```
package package2;

import package1.Parent;

public class Child extends Parent {
```

```
    public void displayChild() {
        System.out.println("Inside Child Class");
        System.out.println("------------------");

        System.out.println("Public Variable    : " + publicVar);
        System.out.println("Protected Variable : " + protectedVar);

        // Not accessible:
        // System.out.println(privateVar);
        // System.out.println(defaultVar);
    }
}
```

File 3: Main.java

```
import package1.Parent;
import package2.Child;

public class Main {

    public static void main(String[] args) {

        Parent parent = new Parent();

        System.out.println("====================================");
        System.out.println("       TINY PROJECT");
        System.out.println("====================================");
        System.out.println();

        System.out.println("Accessing Parent object from Main");
        System.out.println("----------------------------------");

        System.out.println("Public Variable : " + parent.publicVar);

        // Not accessible from another package:
        // System.out.println(parent.privateVar);
        // System.out.println(parent.protectedVar);
        // System.out.println(parent.defaultVar);

        System.out.println();

        Child child = new Child();
        child.displayChild();

        System.out.println();

        parent.display();
    }
}
```

5. Explanation of Access Modifiers

public

A public member can be accessed from any class and any package. In this project, publicVar is accessible from Parent, Child and Main.

private

A private member can be accessed only inside the class where it is declared. privateVar is therefore accessible inside Parent but not directly inside Child or Main.

protected

A protected member is accessible within the same package and by subclasses in another package. Child extends Parent, so Child can access protectedVar.

default

When no access modifier is written, the member has default (package-private) access. It can be accessed within the same package but not from another package.

6. Access Modifier Comparison

Modifier	Same Class	Same Package	Subclass in Different Package	Different Package
public	✓	✓	✓	✓
protected	✓	✓	✓	✗
default	✓	✓	✗	✗
private	✓	✗	✗	✗

7. Expected Output

```
====================================
       TINY PROJECT
====================================

Accessing Parent object from Main
----------------------------------
Public Variable : 10

Inside Child Class
------------------
Public Variable    : 10
Protected Variable : 30

Inside Parent Class
-------------------
Public Variable    : 10
Private Variable   : 20
Protected Variable : 30
Default Variable   : 40
```

8. How to Compile and Run

Compile all source files using:

```
javac -d . src/package1/Parent.java src/package2/Child.java src/Main.java
```

Then run:

```
java Main
```

9. Important Observations

• Main can access the public member of Parent even though Main is outside package1.

• Main cannot directly access private, protected, or default members declared in Parent.

• Child can access publicVar and protectedVar because it extends Parent.

• Child cannot directly access privateVar or defaultVar from package2.

• Parent.display() can access all four members because it runs inside Parent.

10. Conclusion

This tiny project demonstrates Java access control using classes in different packages. Public provides the widest visibility, private provides the most restricted visibility, protected provides controlled access for inheritance and package members, and default limits access to the same package. Access modifiers are an important part of encapsulation and help developers control how class data and behavior are exposed.

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